

Polina Petrov

✉ polina.petrov@vanderbilt.edu polinapetrov.github.io

Education

Vanderbilt University Nashville, TN	Aug 2021 - Present
Ph.D. in Astrophysics; Advisor: Stephen R. Taylor	
Carnegie Mellon University Pittsburgh, PA	2021
B.S. in Physics, Astrophysics Concentration	

Honors & Awards

Graduate

McMinn Research Award	2025
Vanderbilt Graduate School Travel Grant	2024
NASA FINESST Grant	2023 - Present
University Graduate Fellow, Russell G. Hamilton Scholar	2021 - Present

Undergraduate

Mellon College of Science Honors	2021
University Honors	2021
NSF Physics REU	2020

Publications

First-author Publications

- P. Petrov**, S. R. Taylor, M. Charisi, C.P. Ma, "Identifying Host Galaxies of Supermassive Black Hole Binaries Found by Pulsar Timing Arrays", *ApJ*, 976, 129 (2024)
[10.3847/1538-4357/ad7b14](https://doi.org/10.3847/1538-4357/ad7b14)
- P. Petrov**, L. P. Singer, M. W. Coughlin, [and 7 others], "Data-driven expectations for electro-magnetic counterpart searches based on LIGO/Virgo public alerts", *ApJ*, 924, 54 (2022)
[10.3847/1538-4357/ac366d](https://doi.org/10.3847/1538-4357/ac366d)

Co-author Publications

- B. Bécsy, N. Cornish, **P. Petrov**, [and 4 others], "Towards robust gravitational wave detections from individual supermassive black hole binaries" (2025)
[arXiv:2502.18114](https://arxiv.org/abs/2502.18114)

NANOGrav Collaboration Publications

- G. Agazie et al., "The NANOGrav 15 yr Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries", *ApJL*, 951, L50, (2023)
[10.3847/2041-8213/ace18a](https://doi.org/10.3847/2041-8213/ace18a)
- G. Agazie et al., "The NANOGrav 15-year Data Set: Constraints on Supermassive Black Hole Binaries from the Gravitational Wave Background", *ApJL*, 952, L37, (2023)
[10.3847/2041-8213/ace18b](https://doi.org/10.3847/2041-8213/ace18b)
- G. Agazie et al., "The NANOGrav 15-year Data Set: Evidence for a Gravitational-Wave Background", *ApJL*, 951, L8 (2023)
[10.3847/2041-8213/acdac6](https://doi.org/10.3847/2041-8213/acdac6)

Research

Vanderbilt University, Dept of Physics and Astronomy

2021 - Present

Graduate Research, advised by Prof. Stephen R. Taylor

Simulating gravitational wave discoveries of individual supermassive black hole binaries in pulsar timing array data and estimating system parameters using MCMC methods. Developing a framework to quantify and rank host galaxies contained within the GW localization volume, thereby making electromagnetic follow-up more manageable and accelerating multi-messenger discoveries.

Carnegie Mellon University, Dept of Physics

2020 - 2021

Undergraduate Research Assistant to Prof. Tiziana DiMatteo

Compared the frequency-strain evolution of massive black hole binary (MBHB) mergers from the Illustris and IllustrisTNG simulations against the Laser Interferometer Space Antenna (LISA) sensitivity curve to determine the time span during which these systems are detectable. Investigated the effect that MBHB system parameters have on detection time span with LISA.

Global Relay of Observatories Watching Transients Happen (GROWTH)

2020 - 2021

Undergraduate research with Dr. Leo Singer and Prof. Michael Coughlin

Continued Summer 2020 research by simulating detection probabilities of electromagnetic counterparts to gravitational wave sources for future LIGO, Virgo, and KAGRA observing runs O4 and O5. Specifically focused on implications for electromagnetic follow-ups with the Zwicky Transient Facility (ZTF). Results published in ApJ (see previous section).

University of Minnesota, Dept of Physics and Astronomy

Summer 2020

REU Internship, advised by Prof. Michael Coughlin

Optimized the search for electromagnetic observations of gravitational wave sources in collaboration with the GROWTH team. Used ZTF's scheduling software, kilonova light curve simulations, and gravitational wave localization sky maps to compare kilonova detection probabilities of binary neutron star mergers to that of neutron star-black hole mergers.

University of Pittsburgh, Dept of Physics and Astronomy

Summer 2019

Undergraduate Research Assistant to Prof. Carles Badenes

Used sky survey catalogs to study stellar multiplicity statistics. Examined for red giant stars the relationship between maximum shifts in radial velocity and excess ultraviolet and optical radiation expected in the presence of a companion star. Compared survey data to broadband spectral energy distributions of APOGEE-2 sources fitted using the MIST models and Kurucz stellar atmosphere models.

Presentations

"Detection Milestones for Individual Binaries Detected by PTAs"

Oral presentation, IPTA Collaboration Meeting | Pasadena, CA

June 2025

"Identifying Host Galaxies of Supermassive Black Hole Binaries Found by PTAs"

Oral presentation, The Era of Binary Supermassive Black Holes Meeting | Aspen, CO

Feb 2025

Oral presentation, Astronomy Dept Lunch Talk | University of California, Berkeley

Dec 2024

Oral presentation, IPTA Collaboration Meeting | Sexten, Italy

June 2024

"Optimizing Host Galaxy Identification of Individual Supermassive Black Hole Binaries"

Oral presentation, NANOGrav Spring Meeting | Zoom (virtual)

Mar 2024

Oral presentation, AAS January Meeting | New Orleans, LA

Jan 2024

Oral presentation, NANOGrav Fall Meeting | University of British Columbia

Oct 2023

"Mapping the Host Galaxies of Nanohertz Gravitational Wave Sources"

Oral presentation, IPTA Collaboration Meeting | Zoom (virtual)

June 2023

Oral presentation, APS April Meeting | Minneapolis, MN

Apr 2023

Oral presentation, NANOGrav Fall Meeting | University of Wisconsin-Milwaukee

Oct 2022

Poster presentation, GMT Meeting: Black Holes at All Scales | Sedona, AZ

Sept 2022

Oral presentation, VIPER Summer School on PTA GW Astrophysics | Nashville, TN

July 2022

Teaching

Graduate Teaching Assistant | Vanderbilt University

ASTR 1010L: Introductory Nighttime Astronomy Laboratory

2021 - 2023

ASTR 1020L: Introductory Daytime Astronomy Laboratory

Fall 2022

Mentoring & Outreach

Research mentor to undergraduate student Tejas Mahesh | Vanderbilt University

June 2025 - Present

NANOGrav Student Workshop Organizing Committee | Ann Arbor, MI

Oct 2024

VIPER Summer School tutorial leader | Vanderbilt University

July 2024

Research mentor to undergraduate student Celine Mang | Vanderbilt University

Spring 2024

Astronomy on Tap organizer | Nashville, TN

2023 - Present

2022 Physics Concepts mentor | Carnegie Mellon University

2018

Buhl Planetarium volunteer | Pittsburgh, PA

2016 - 2018

Professional & Academic Affiliations

NANOGrav Full Member

2024 - Present

NANOGrav Associate Member

2021 - 2024

Sigma Pi Sigma Honor Society

2021 - Present

American Astronomical Society (AAS)

2017 - Present

American Physical Society (APS)

2017 - Present